

IT3051 - FUNDAMENTALS OF DATA MINING

FINAL TECHNICAL REPORT

Hotel Booking Cancellation Prediction Using Machine Learning

Project Information	Details
Module	IT3051 - Fundamentals of Data Mining
Project Type	Group Mini Project - Technical Report
Machine Learning Task	Supervised Binary Classification
Dataset	Hotel Booking Demand
Target Variable	is_canceled
Group Members	Gowsika Surendran (IT23794566) Manoharan Subasthican (IT23810778) Kuganesan Kageepan (IT23717572) Kokulan Seyon (IT23752672)
Academic Year	2026

BSc (Hons) in Information Technology - Data Science

Executive Summary

This project develops a machine-learning-based Hotel Booking Cancellation Prediction System using the Hotel Booking Demand dataset. The business problem is the uncertainty created by reservation cancellations, which can affect room planning, occupancy forecasting, resource allocation and revenue management. The prediction objective is to classify a booking as cancelled or not cancelled using relevant booking information available before the final reservation outcome is known.

The original dataset contains 119,390 booking records and 32 columns from a City Hotel and a Resort Hotel, with arrival dates from July 2015 to August 2017. The target variable is `is_canceled`, where 0 represents a booking that was not cancelled and 1 represents a cancelled booking. The original target distribution is 75,166 not-cancelled bookings (62.96%) and 44,224 cancelled bookings (37.04%), indicating moderate class imbalance.

During data understanding and preprocessing, missing values, duplicate records, unusual numerical observations, categorical variables and data-leakage risks were investigated. A total of 31,994 duplicate rows were identified and removed, leaving 87,396 rows. Missing values were found in `children`, `country`, `agent` and `company`. The preprocessing workflow used justified missing-value treatment, feature engineering, leakage removal, train-test separation, standardization of numerical variables and one-hot encoding of categorical variables.

FINAL RESULT REQUIRED: Insert the final four-model comparison, tuned-model results, selected final model, system test results and final business recommendation after Stages 6-10 are completed.

Table of Contents

Update the automatic Table of Contents in Microsoft Word before submission: References > Table of Contents > Automatic Table. The report structure below follows Stage 11 of the IT3051 Mini Project Assignment.

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1. Introduction and Problem Definition

1.1 Background

Hotel reservations are affected by uncertainty because some customers cancel before arrival. Cancellations can create operational and financial challenges for hotels by changing expected occupancy, room availability, staffing requirements and revenue forecasts. A data-driven cancellation prediction system can support hotel staff by identifying bookings that show patterns associated with cancellation risk.

1.2 Problem Statement

The problem addressed in this project is to predict whether an individual hotel reservation is likely to be cancelled. The system is designed as a decision-support tool rather than an automated decision-maker. It uses historical booking characteristics to generate a cancellation classification that can support reservation planning.

1.3 Project Objective

The main objective is to develop a complete data mining and machine learning solution that analyses hotel booking information and predicts whether a reservation is likely to be cancelled.

- Understand and prepare the Hotel Booking Demand dataset.
- Identify important data-quality issues and booking patterns through EDA.
- Engineer and select suitable predictive features while preventing data leakage.
- Develop and compare at least four suitable classification algorithms.
- Optimize promising models and select a final model using experimental evidence.
- Integrate the final preprocessing pipeline and model into a usable prediction system.
- Present prediction results in a clear form that can support hotel reservation decisions.

1.4 Prediction Task

The target variable is `is_canceled`. A value of 0 means the booking was not cancelled and a value of 1 means the booking was cancelled. Because the target contains two discrete outcomes, the project is a supervised binary classification task.

2. Scenario, Stakeholders and User Requirements

2.1 Intended Stakeholders

- Hotel managers - to understand cancellation risk and support operational planning.
- Reservation staff - to review booking risk information when managing reservations.
- Revenue-management staff - to support occupancy and revenue planning.
- Operations planners - to anticipate changes in room and resource requirements.

2.2 User Requirements

- The system should accept relevant booking information using clear input fields.
- Inputs should be validated before a prediction is generated.
- The system should automatically apply the same feature engineering and preprocessing used during model development.
- Outcome-related leakage variables must not be requested as predictive inputs.
- The result should clearly indicate the predicted cancellation class.
- The interface should be understandable to a non-technical hotel user.

2.3 Expected Inputs and Output

Potential user inputs include hotel type, lead time, arrival-date information, stay duration, guest composition, meal plan, country, market segment, distribution channel, repeated-guest status,

previous booking behaviour, room type, deposit type, customer type, ADR, parking requirements and special requests. Engineered features are calculated automatically. The output is a prediction of 'Likely to Cancel' or 'Not Likely to Cancel'.

3. Dataset Identification, Source and Validation

3.1 Dataset

Attribute	Description
Dataset name	Hotel Booking Demand
Original records	119,390
Original columns	32
Hotel types	City Hotel and Resort Hotel
Arrival period	July 2015 - August 2017
Target	is_canceled
Task	Binary classification

3.2 Source and Citation

The dataset is publicly distributed through Kaggle and originates from the research publication by Antonio, de Almeida and Nunes (2019), 'Hotel booking demand datasets', Data in Brief, 22, 41-49.

Dataset source: Kaggle - Hotel Booking Demand (Jesse Mostipak distribution).

3.3 Suitability

The dataset is suitable because it has a clear binary target, a large number of labelled records, mixed numerical and categorical variables, realistic data-quality issues and sufficient complexity for EDA, preprocessing, feature engineering, model comparison and optimization. It also represents a clear real-world decision-support scenario.

3.4 Validation

The Hotel Booking Demand dataset was submitted for instructor validation and approved before the group proceeded with EDA and preprocessing.

4. Data Understanding and Exploratory Data Analysis

4.1 Dataset Structure and Target Distribution

Target class	Count	Percentage
Not Cancelled (0)	75,166	62.96%
Cancelled (1)	44,224	37.04%

The target is moderately imbalanced. The imbalance is not extreme, but model evaluation should not rely on accuracy alone. Precision, recall, F1-score, confusion matrix and ROC-AUC are therefore appropriate later evaluation measures.

4.2 Missing Values

Feature	Missing values	Observation
children	4	Very small amount of missingness.
country	488	Missing country category requires treatment.
agent	16,340	Approximately 13.69% missing.
company	112,593	Approximately 94.31% missing; requires careful interpretation.

4.3 Duplicate Records

The dataset contained 31,994 exact duplicate rows. After duplicate removal, 87,396 rows remained. Duplicate removal was performed before the later modelling workflow.

4.4 Outliers and Unusual Observations

Variable	Observed range / unusual value	Interpretation
ADR	-6.38 to 5400	Contains negative and very high values that require investigation.
Lead time	Maximum 737 days	Very long advance bookings exist.
Total nights	Maximum 69 nights	Long-stay observations are present.
Days in waiting list	Maximum 391 days	Highly concentrated/skewed with extreme values.

The PE1 notebook investigated these unusual values but did not apply a separate outlier-removal or capping step. Therefore, the report distinguishes outlier investigation from actual outlier treatment.

4.5 Important EDA Findings

Analysis	Observed result	Interpretation
Hotel type	City Hotel 41.73% cancellation; Resort Hotel 27.76%	Cancellation behaviour differs by hotel type.
Deposit type	Non Refund 99.36%; No Deposit 28.38%; Refundable 22.22%	Deposit type has a strong observed association with cancellation.

Customer type	Transient 40.75%; Contract 30.96%; Transient-Party 25.43%; Group 10.23%	Cancellation rates vary by customer type.
Special requests	Not cancelled mean 0.714; cancelled mean 0.329	Non-cancelled bookings had more special requests on average.
Market segment	Groups 61.06%; Online TA 36.72%; Offline TA/TO 34.32%; Direct 15.34%	Cancellation patterns differ by market segment.

These are associations observed in the dataset and should not be interpreted as proof that the variables cause cancellation.

4.6 Correlation Analysis

A numerical correlation heatmap was produced to inspect linear relationships between numerical variables. Correlation was used as an exploratory tool only; correlation does not imply causation. Exact coefficients should be reported from the final analysis output where required.

4.7 Data Leakage Analysis

reservation_status and reservation_status_date were identified as clear outcome-related leakage variables for a pre-outcome cancellation prediction scenario. These fields were removed from the predictive feature set. Timing-sensitive variables should also be interpreted according to whether they would genuinely be available at the intended prediction time.

5. Data Cleaning and Preprocessing

5.1 Duplicate Removal

Exact duplicate rows were removed using duplicate detection and removal, reducing the data from 119,390 to 87,396 records.

5.2 Missing-Value Treatment

Feature	Treatment	Reason / interpretation
children	Median imputation	Only four values were missing; median provides a robust central value.
country	'Unknown' category	Preserves the booking while explicitly representing missing country information.
agent	0	Used in the implementation as a missing/no-record representation.
company	0	Used in the implementation as a missing/no-record representation; not interpreted as a real company ID.

5.3 Categorical Encoding

Categorical variables were transformed using `OneHotEncoder(handle_unknown='ignore')`. One-hot encoding converts nominal categories into binary indicator columns without introducing an artificial order. The `handle_unknown` option allows unseen categories in later data to be processed without failing.

5.4 Numerical Scaling

Numerical variables were processed using median imputation within the modelling pipeline and `StandardScaler`. Standardization is particularly useful for algorithms that are sensitive to differences in feature magnitude.

5.5 Train-Test Split and Leakage Prevention

Dataset	Rows
Training set	69,916
Test set	17,480

An 80:20 train-test split was used with `random_state=42` for reproducibility and `stratify=y` to preserve the target-class proportions. The preprocessing transformer was fitted on `X_train` and then used to transform `X_test`, reducing the risk of test-set information leaking into training.

After preprocessing, the training and test matrices contained 254 processed features because one-hot encoding expanded categorical variables into multiple binary columns.

6. Feature Engineering and Feature Selection

6.1 Engineered Features

Feature	Definition	Purpose
<code>total_nights</code>	<code>stays_in_weekend_nights + stays_in_week_nights</code>	Represents total planned stay duration.
<code>is_long_stay</code>	1 if <code>total_nights > 7</code> , otherwise 0	Flags bookings with longer stays.
<code>is_weekend_stay</code>	1 if weekend nights <code>> 0</code> , otherwise 0	Flags bookings that include a weekend.
<code>is_high_lead_time</code>	1 if <code>lead_time > 60</code> , otherwise 0	Flags bookings made relatively far in advance.

6.2 Feature Selection

Feature selection considered relevance to the cancellation scenario, data quality, redundancy, prediction-time availability and leakage risk. `reservation_status` and `reservation_status_date` were excluded because they reveal outcome-related information. Other features should be retained or removed only with documented justification.

7. Model Development and Algorithm Rationale

The assignment requires at least four suitable machine learning algorithms to be implemented and compared. The following four models are suitable for the binary classification problem:

Algorithm	Rationale
Logistic Regression	Provides an interpretable linear baseline and probability-based binary classification.
Decision Tree	Captures non-linear rules and interactions and is relatively easy to interpret.
Random Forest	Combines multiple decision trees to improve robustness and reduce overfitting compared with a single tree.
Gradient Boosting	Builds an ensemble sequentially to correct earlier errors and can model complex non-linear relationships.

FINAL RESULT REQUIRED: Insert the exact implementation settings for all four baseline models from the final modelling notebook.

8. Model Evaluation and Comparison

Models should be evaluated on held-out data using metrics appropriate to binary classification. Because the target is moderately imbalanced, accuracy alone is insufficient.

Metric	Purpose
Accuracy	Overall proportion of correct predictions.
Precision	Among bookings predicted as cancelled, the proportion that were actually cancelled.
Recall	Among actually cancelled bookings, the proportion correctly identified.
F1-score	Balances precision and recall.
Confusion Matrix	Shows true/false positives and true/false negatives.
ROC-AUC	Measures ranking/discrimination performance across classification thresholds.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	PENDING	PENDING	PENDING	PENDING	PENDING
Decision Tree	PENDING	PENDING	PENDING	PENDING	PENDING
Random Forest	PENDING	PENDING	PENDING	PENDING	PENDING
Gradient Boosting	PENDING	PENDING	PENDING	PENDING	PENDING

FINAL RESULT REQUIRED: Replace PENDING values with the actual test/validation results. Add confusion matrices and ROC curves from the final experiments. Do not select a winner before these results exist.

9. Hyperparameter Tuning and Optimization

Promising baseline models should be tuned using an appropriate search strategy such as GridSearchCV or RandomizedSearchCV with cross-validation. The purpose is to test whether better parameter combinations improve generalization rather than simply improving training performance.

Model	Example parameters to investigate
Decision Tree	max_depth, min_samples_split, min_samples_leaf, criterion
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features
Gradient Boosting	n_estimators, learning_rate, max_depth / base-tree complexity, subsample
Logistic Regression	C, penalty, solver where compatible

FINAL RESULT REQUIRED: Insert the actual tuning method, search space, cross-validation setup, best parameters and baseline-vs-tuned results.

10. Final Model Selection and Justification

The final model must be selected using experimental evidence rather than preference. Selection should consider the relevant evaluation metrics, stability across validation, generalization to the test set, interpretability, computational requirements and suitability for integration into the final prediction system.

FINAL RESULT REQUIRED: State the selected final model and justify it using the actual baseline/tuned comparison. Include exact metrics and explain any trade-offs between precision, recall, F1-score and ROC-AUC.

11. System Architecture and Implementation

The proposed end-to-end architecture is:

User Input -> Validation -> Feature Engineering -> Preprocessing Pipeline -> Final Model -> Prediction Result

11.1 Prediction Workflow

1. The user enters booking information through the frontend.

2. The system validates required values and acceptable ranges/categories.
3. Derived features such as `total_nights` and indicator features are generated automatically.
4. The saved preprocessing pipeline applies the same transformations used during model development.
5. The final trained model generates the cancellation prediction.
6. The system displays a clear result such as Likely to Cancel or Not Likely to Cancel.

11.2 Model and Pipeline Persistence

The final trained model and fitted preprocessing pipeline should be saved and loaded by the backend. This ensures that live user inputs receive exactly the same preprocessing logic as the data used during model development.

FINAL RESULT REQUIRED: Insert the final technology stack, saved model/pipeline filenames and an architecture diagram or screenshot of the implemented system.

12. Backend and Frontend Development

12.1 Backend

The backend should receive validated inputs, create the engineered features, apply the fitted preprocessing pipeline, call the final model and return a meaningful prediction. It should also handle missing or invalid inputs safely.

12.2 Frontend

The frontend should provide clear input fields, appropriate dropdowns or numerical controls, validation messages and a readable prediction result. The interface should avoid requesting outcome-related variables such as `is_canceled`, `reservation_status` or `reservation_status_date`.

FINAL RESULT REQUIRED: Insert screenshots of the final frontend, describe the framework used (for example Streamlit if chosen), and document the input validation rules.

13. System Testing and Results

End-to-end testing should confirm that the frontend and backend work together, valid inputs generate predictions, invalid inputs are handled correctly, engineered features are created consistently and the saved preprocessing/model pipeline produces the expected output.

Test case	Expected result	Actual result	Status
Valid normal booking	Prediction returned	PENDING	PENDING
Missing required input	Validation message	PENDING	PENDING
Invalid numeric value	Validation message	PENDING	PENDING
Unseen categorical value	Handled by preprocessing	PENDING	PENDING
Known test booking	Consistent model output	PENDING	PENDING

FINAL RESULT REQUIRED: Complete the test table using the implemented application and include representative screenshots.

14. Limitations and Possible Future Improvements

- The dataset represents two hotels and historical arrivals from 2015-2017; results should not automatically be generalized to every hotel or current market condition.
- Some fields may be available only at particular stages of the booking lifecycle, so the production prediction point must be defined carefully.
- The company field has very high missingness, which limits how confidently it can be interpreted.
- Unusual numerical values such as extreme ADR and long lead times require careful business interpretation.
- Observed EDA relationships are associations and do not establish causation.
- Model performance can change over time as customer behaviour and booking channels change.

14.1 Future Improvements

- Validate the model on newer data and additional hotels.
- Perform more detailed outlier validation with domain experts.
- Investigate feature importance or explainability methods for the selected final model.
- Evaluate probability calibration if cancellation probabilities are shown to users.
- Monitor model performance after deployment and retrain when data distributions change.
- Improve the interface with clearer explanations and accessible visual feedback.

14.2 Ethical and Responsible Use

The prediction should be used as decision-support information and should not be the sole basis for denying service or unfairly treating customers. Potential differences across hotel types, countries, market segments or customer groups should be communicated as limitations rather than presented as universal behaviour.

15. Individual and Group Contributions

The group should update this table so that it accurately reflects the work actually completed by each member across the full project. Every member should understand the complete workflow because the assignment includes individual evaluations.

Member	Primary contribution areas
Gowsika Surendran (IT23794566)	Data understanding, missing values, target distribution, duplicate analysis; contribute to modelling/report/presentation.
Manoharan Subasthican (IT23810778)	EDA of hotel/customer/market/month/channel relationships; contribute to modelling/report/presentation.

Kuganesan Kageepan (IT23717572)	EDA of lead time, ADR, stay length, room behaviour and special requests; contribute to modelling/report/presentation.
Kokulan Seyon (IT23752672)	Country/high-cardinality analysis, leakage assessment and preprocessing decisions; contribute to modelling/report/presentation.

FINAL RESULT REQUIRED: Revise these contributions after model development, tuning, backend/frontend work, testing and final report preparation so the table reflects the actual full-project contribution.

16. Conclusion

The Hotel Booking Cancellation Prediction project applies the complete data mining workflow to a real-world hotel reservation problem. The approved Hotel Booking Demand dataset provides a clear binary target and sufficient complexity for meaningful EDA, preprocessing, feature engineering and model development. The completed PE1 work identified missing values, duplicate records, unusual observations, class imbalance, important cancellation patterns and leakage risks. The preprocessing workflow removed exact duplicates, treated missing values, created four engineered features, excluded outcome-related leakage variables, performed a stratified train-test split, standardized numerical variables and one-hot encoded categorical variables.

The final conclusion must be updated after model development, optimization and system testing to report the selected model, its verified performance, the implemented application and the evidence-based business value of the completed solution.

References

Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41-49. <https://doi.org/10.1016/j.dib.2018.11.126>

Kaggle. Hotel Booking Demand. Dataset distribution by Jesse Mostipak.
<https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>

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Appendix A - Final Submission Checklist

Requirement	Status
Approved dataset and source/citation	Available
Dataset proposal/validation evidence	Available
EDA and preprocessing source code	Available
At least four ML algorithms	TO COMPLETE

Model comparison/evaluation evidence	TO COMPLETE
Hyperparameter tuning	TO COMPLETE
Trained final model	TO COMPLETE
Integrated backend/frontend source code	TO COMPLETE
System testing evidence	TO COMPLETE
Technical report	Draft created - update final results
Final presentation materials	TO COMPLETE

Appendix B - Technical Report Mapping to Assignment

Assignment Stage 11 requirement	Report section
Introduction and problem definition	Sections 1
Scenario and user/stakeholder requirements	Section 2
Dataset identification, source, citation and validation	Section 3
Data understanding and EDA	Section 4
Data cleaning and preprocessing	Section 5
Feature engineering and feature selection	Section 6
Algorithms implemented and rationale	Section 7
Model evaluation and comparison	Section 8
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